

Study of Total Column Ozone and Direct Solar Irradiance (UV-B) at Maitri, Antarctica

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ABSTRACT

During the 16th, 21st, 22nd and 23rd Indian Antarctic expedition conducted in 1997, 2002-03 and 2004 respectively, measurement of columnar ozone and direct UV irradiances have been made using MICROTOPS II Sun-photometer from the Indian station Maitri (70° 45' S, 11° 44' E). It is found that the ozone hole depth fluctuated from year-to-year. The minimum columnar ozone observed over Maitri was 135 (± 9.3) DU, 185 (± 12) DU, 126 (± 9.7) DU and 159.8 (± 8.8) DU during spring 1997, 2002, 2003 and 2004, respectively. Ozone concentration below 220 DU (nominal ozone threshold value) during Antarctic spring (day 225–365) over Maitri has been observed to be 45.1 %, 20.7 %, 62.7 % and 60 % in 1997, 2002, 2003 and 2004, respectively. The instantaneous irradiances at 305 ± 0.9 , 312 ± 0.9 and 320 ± 1.0 nm during a no-ozone-hole (13 December 2004) and an ozone-hole (4 October 2004) period have been observed to be about 0.031, 0.150 and 0.299 W m⁻² and 0.010, 0.049 and 0.102 W m⁻², respectively at local noontime. The observations of the direct UV irradiances are compared with tropospheric ultraviolet visible (TUV) radiation transfer model calculations. The model estimate shows that, during the ozone-hole period, a loss of ozone of the order of 44% leads to an increase in irradiance of the order of 410%, 90% and 25% at 305 ± 0.9 , 312 ± 0.9 and 320 ± 1.0 nm, respectively. The relationship between change in UV irradiance due to a change in column ozone is obtained using a TUV model and irradiances thus estimated from this relationship are found to compare well with the observed irradiances.

Keywords: Antarctica, Ozone hole, UV-B irradiance, TUV model

INTRODUCTION

Total Ozone Column (TOC) is the total amount of ozone present in a column extending vertically from Earth's surface to the top of the atmosphere. Stratospheric ozone protects life on the surface of the Earth, by screening out genetically destructive ultraviolet radiation from Sun and shielding the Earth surface from harmful ultraviolet radiation. Total Ozone Column (TOC) is photo-chemically controlled by chemically active species in the oxygen, hydrogen, nitrogen, chlorine and bromine families. Most of the ozone is produced over the tropics and transported into the polar region by the Brewer-Dobson circulation. The total ozone column is decreasing at mid latitudes of northern and southern hemisphere (Harris et al, 1997). In 1985, Farman et al of the British Antarctic Survey first reported the depletion of ozone over Antarctica during the southern hemisphere spring period. Thereafter, numerous studies based on various ground and satellite measurements (Crutzen and Arnold, 1986; Manney et al, 1996; Kondratyev and Varotsos, 2000; Johnston, 1971; Farman et al, 1985; Hartmann, 1994; Newman, 1986) have shown an overall declining trend of total ozone (TOZ) content globally during 1979-94.

The formation of ill-famed ozone hole over Antarctica during Antarctic spring is mainly because of catalytic depletion of ozone due to heterogeneous chemical reactions which take place on the surface of the polar stratospheric cloud (PSCs) (Crutzen and Arnold, 1986; Manney et al, 1996; Kondratyev and Varotsos, 2000; Johnston, 1971) inside the polar vortex. Two types of PSCs have been discovered, (Crutzen and Arnold, 1986) that provide a surface for heterogeneous reactions to take place (Cadle et al, 1975) in the presence of sunlight, which enhance the production and lifetime of reactive chlorine that convert less reactive molecules into much more reactive forms which readily destroy ozone (WMO, 1999).

Ground-based measurements, satellite estimates and model calculations of ultraviolet irradiance, particularly in the UV-B range, have recently been reported and helped in quantifying effects of solar ultraviolet radiation on biological systems (McKinlay and Diffey, 1987; Frederick and Lubin, 1988; Cullen et al, 1992; Bais et al, 1993; Zerefos et al, 1995; Bernhard et al, 1997; Herman et al, 1996; Katsambas et al, 1997; Minschwaner, 1999; Madronich et al, 1998; Feretis et al, 2002; Palancar and Toseli, 2002). The evidence is overwhelming that under cloud-free skies UV-B radiation is controlled largely by ozone (Bais et al, 1993; Kondratyev et al, 1995; Bodhaine et al, 1997;

Mayer et al, 1997; Taalas et al, 1997; Varotsos et al, 2001). The response of UV-B radiation to ozone changes is strongly dependent on wavelength because of the rapid increase of the ozone absorption cross section toward shorter wavelengths, with greater sensitivity at shorter wavelengths and low sun, where the slant ozone optical depth is greater (Madronich et al, 1995; Varotsos et al, 1995; Varotsos et al, 1995a; Fioletov and Evans, 1997). Ozone depletion specifically enhances the shorter wavelengths of the UV spectrum (280-320 nm) and allows more biologically damaging, UV-B radiation to reach the earth's surface (Briehl and Crutzen, 1989; Kondratyev and Varotsos, 1996; Kondratyev and Varotsos, 1996a; Taalas et al, 1997; Udelhofen et al, 1999; McKenzie et al, 2001). It is reported that a 1% decrease in stratospheric ozone could cause about 2% increase in UV-B radiation (Cutchis, 1974; Efsthathiou et al, 1998; Varotsos, 2005). This increase may vary depending on specific wavelength, season and solar zenith angle (Bais et al, 1993; Eck et al, 1995; Varotsos et al, 1998; Alexandris et al, 1999). High levels of UV-B radiation have been observed directly in association with the Antarctic ozone hole (Frederick and Alberts, 1991; Booth et al, 1994; Johnston et al, 1995; Frederick et al, 1998). At several occasions the measured DNA-damaging radiation at Palmer Station, Antarctica (64°S) has been found to exceed maximum summer values at San Diego, USA (32°N) (Booth et al, 1994). It should be noted that monitoring of UV-B irradiances at Maitri began in mid 1980's, well after the appearance of the ozone hole; hence the UV-B levels in pre-ozone-hole years can be only estimated.

In view of the above, a hand held microprocessor based MICROTUPS II Sunphotometer, had been used to measure total column ozone and UV irradiance at Maitri, Indian Antarctica research station during the years 1997, 2002 – 04. More details about the MICROTUPS instrument can be found elsewhere (Morys et al. 2001). The observations were taken throughout the expedition on hourly basis during clear sunny days. The observed TOC was compared with satellite (TOMS) data. Monthly averaged temperature at 30hPa, 50hPa and 100hPa was compared with TOC observed in 1997, 2002-2004. Similarly, in order to study the UV irradiance and to evaluate the measured UV irradiances under clear sky conditions, we have also made use of Tropospheric Ultraviolet and Visible (TUV) Radiation Model calculations.

OBSERVATIONS AND RESULTS

Annual Variation of Total Column Ozone at Maitri

Variation of columnar ozone as observed by Microtops sunphotometer over Maitri and Novolazarevskaya, Russian Antarctic Station (at two and half km southeast of Indian Station), during the year 1997, 2002, 2003 and 2004 is shown in Figure 1(a) and 1(b) respectively.

Similarly, change in total column ozone values specifically during the ozone hole period as observed by Microtops sunphotometer over Maitri and Novo, in the year 1997, 2002, 2003 and 2004 is shown in Figure 2(a) and 2(b) respectively. The gap in the Figure 1(a) and 1(b) is due to polar night period. Similarly, the gaps in Figure 2 (a) and 2(b) are due to cloudy days on which Sunphotometer could not be used to make observations.

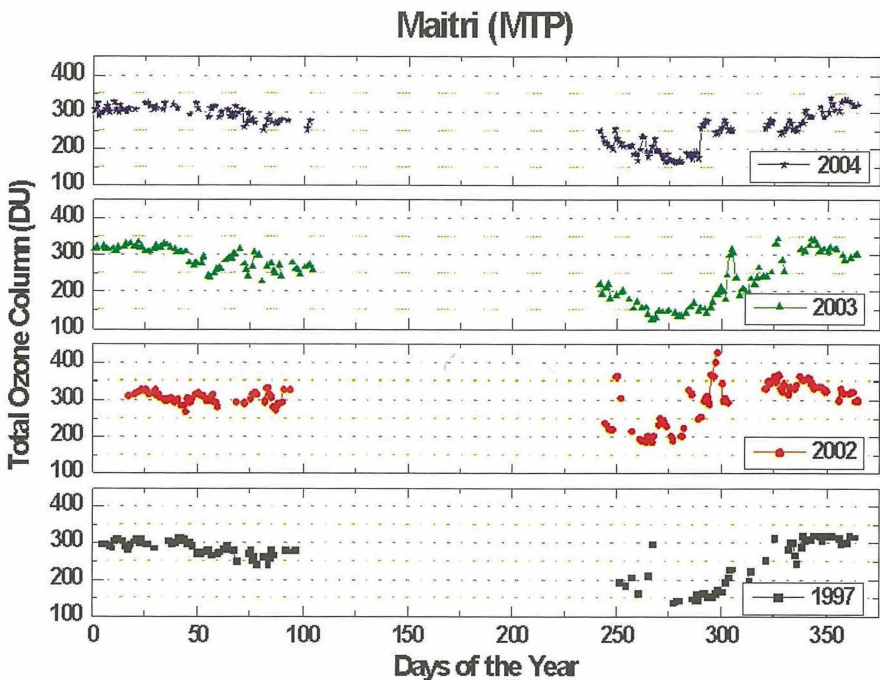


Fig. 1 (a): Variation of columnar ozone observed by Microtops sunphotometer over Maitri during the year 1997, 2002, 2003 and 2004

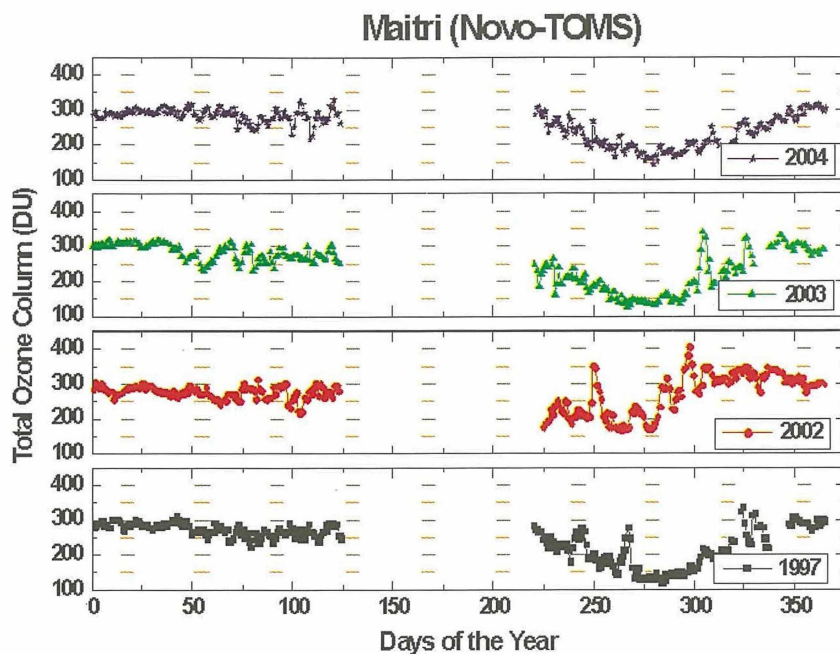


Fig. 1 (b): Variation of columnar ozone observed by TOMS over Maitri (Novo, Russian Antarctic Station), during the year 1997, 2002, 2003 and 2004

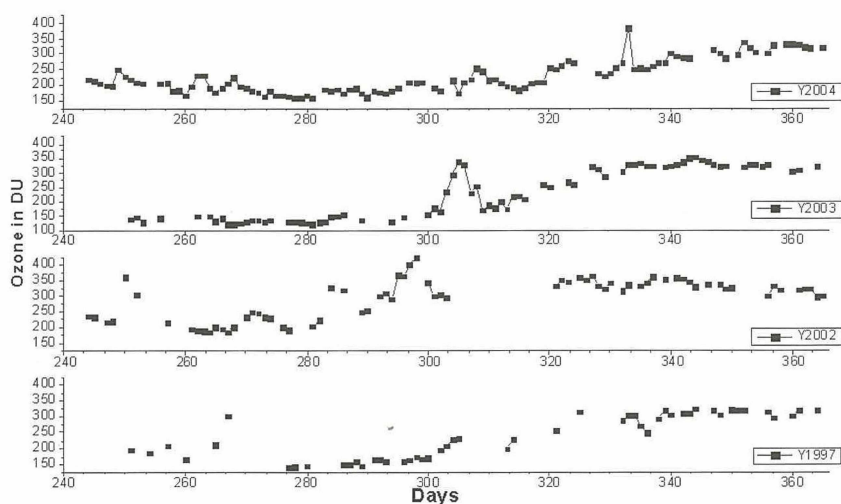


Fig. 2 (a): Comparison of column ozone over Maitri during ozone hole period of 1997, 2002, 2003 and 2004.

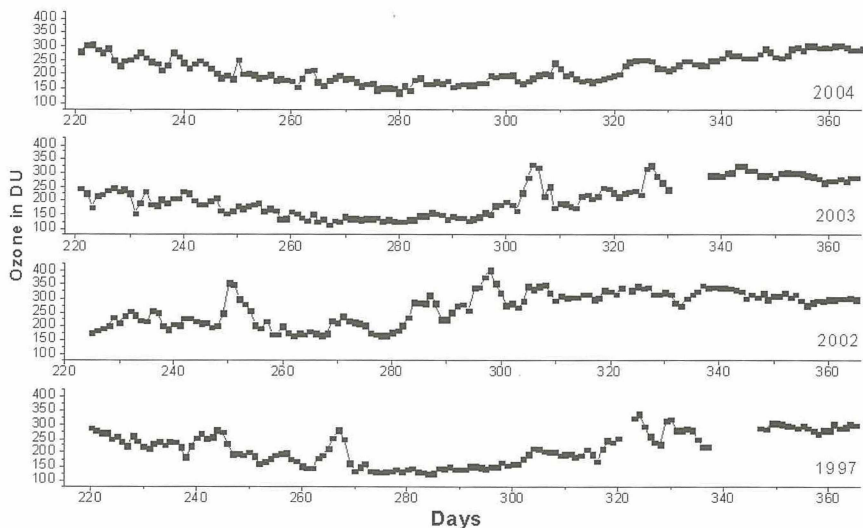


Fig. 2(b): Total column ozone observed by TOMS instrument aboard Earth Probe satellite during ozone hole period over Russian station, Novolazarevskaya

Total ozone up to 320 DU has been observed over Maitri in the months of January and February during all the observational years. The minimum total ozone of about $135 (\pm 9.3)$ DU on 25 September 1997, $185 (\pm 12)$ DU on 20 September 2002, $126 (\pm 9.7)$ DU on 3 October 2003 and $159.8 (\pm 8.8)$ DU on 5 October 2004 has been recorded over Maitri during ozone hole period. These observations showed that the depth of ozone hole varied considerably from year to year while the deepest ozone hole was observed in 2003 compared to 1997, 2002 and 2004. The minimum total ozone of about 122 DU on 11 October 1997, 165 DU on 19 September 2002, 114 DU on 24 September 2003 and 137 DU on 6 October 2004 has been observed by TOMS instrument during ozone hole period. Like Maitri, variation of total ozone has also been observed over Russian station. Both the measurements showed the prevalence of annual ozone hole during all the observational years. However, year-to-year variation in the depth and duration of the ozone hole is seen over Schirmacher region of east Antarctica. Occasionally large fluctuation in total ozone values has also been observed either during the ozone hole period or during recovery period during 1997, 2002 and 2003. However, ozone hole during 2004 do not show such large fluctuation over the Schirmacher region.

The day-to-day fluctuation of columnar ozone during the ozone hole period is clearly seen in Figure 3, which displays the inter-sequential change of the daily ozone values during 1997, 2002, 2003 and 2004. The inter-sequential change is calculated by subtracting total ozone values at successive days ($y_{i+1} - y_i$). As seen from Figure 3, the large fluctuations in day-to-day ozone were observed during 2002. Also, ozone hole during 1997 showed similar fluctuation during ozone hole period; however magnitude of fluctuation was less during deep ozone hole period (days 275-315). Figure 3 further illustrates large day-to-day fluctuations of total ozone over Schirmacher region during 1997, 2002 and 2003 compared to 2004. The main conclusion drawn from the Figure 3 is that the factors that influence the chemical ozone destruction were quite variable year-to-year. This indicates that the stability of ozone hole may largely be controlled by the dynamical behavior of the factors that influences the chemical ozone destruction.

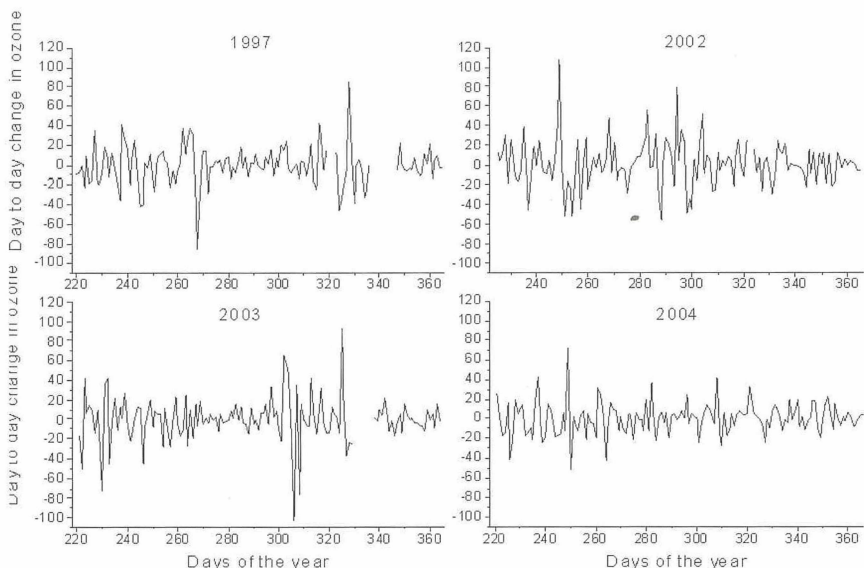


Fig. 3: Day-to-day change of daily total ozone during ozone hole period of 1997, 2002, 2003 and 2004.

As seen from Figure 1, ozone hole during 2002 recovered earlier than recovery observed in 1997, 2003 and 2004. Exceptionally high columnar ozone has been observed during the ozone hole period of 1997 and 2002. Ozone hole during 1997 was quite deep as compared to 2002.

Columnar ozone up to 372 DU on 7 - 8 September 2002 and 300 DU on 23 September 1997 has been observed during ozone hole period. During 2002, observation at Maitri showed recovery around 24 – 25 September and columnar ozone temporarily shoots up to 420 DU in late October during recovery period. The abrupt anomalously high columnar ozone observed in 1997 and 2002 may be attributed to enhanced advection of ozone rich air into the polar vortex during ozone hole period which causes the unstable polar vortex condition over Schirmacher region.

The unstable polar vortex condition observed over Schirmacher region in 1997 and 2002 may be attributed to planetary wave phenomenon (Jain et al, 2004). The Schirmacher region is situated in the fringe region of the vortex. The planetary wave phenomenon forces the ozone rich air masses from mid latitude to polar latitude region and the rapid displacement of the polar vortex from a roughly symmetric circulation about the pole to a circulation that is offset from the pole. If the strength of these waves is high enough then this forces the ozone rich air mass into the polar vortex and sudden increase in the total ozone may takes place. Varotsos (2002) have performed a Fourier analysis of the 10-hpa height and the temperature time series at the high latitude of the southern hemisphere. He found that extremely large amplitude of planetary waves were present which broke up Antarctic ozone hole into two holes (Varotsos, 2003b) on 24-25 September, 2002. After the breakup, the polar vortex reformed and subsequently disappeared very early. This indicates that during the reformation of polar vortex, planetary wave activity induced stratospheric warming leads to weakened potential vorticity and enhanced ozone content in the fringe region of vortex, due to horizontal mixing of air mass from lower latitude, and therefore, filling the vortex with ozone rich air mass.

To find out the distribution of total ozone values over Schirmacher region during ozone hole period, frequency distribution analysis over August to December period has been done against the concentration interval of 0-135 DU, 0-150 DU, 150-220 DU and 220 DU and above, as depicted in Table 1. Total ozone data were available for 135 days in 1997, 140 days in 2002, 138 days in 2003 and 125 days in 2004 during ozone hole period (August to December). Out of these observational days, ozone concentration less than 150 DU has been observed 8.1 %, 0 %, 15 % and 3.2 % of days in 1997, 2002, 2003 and 2004 respectively. It appears that the ozone hole in 2003 was much deeper as compared to 1997, 2002 and 2004. Similarly, ozone concentration below 220 DU (nominal ozone threshold value) during

Antarctic spring has been observed on 45.1 %, 20.7 %, 62.7 % and 60 % of days in 1997, 2002, 2003 and 2004 respectively. Table 1 illustrates the chemical loss of ozone (total ozone concentration below 220 DU) over Schirmacher region during 2003 (day 225 – 365) was increased by a factor of 0.4 compared to 1997 and by a factor of 2 compared to 2002. Out of 140 days, not more than 20.7 % showed ozone concentration below 220 DU and 0% showed ozone concentration below 150 DU observed during 2002, which indicates that the chemical ozone destruction inside the vortex was significantly less during 2002.

Table 1 : Frequency distribution of ozone values during ozone hole period (August to December) observed over the Maitri by TOMS

Year	Days of observation (Aug. to Dec.)	Values less than 135DU	Values less than 150DU	Values between 150-220 DU	Values Above 220 DU	Days of ozone hole in %
1997	135	1 (0.74%)	11 (8.1%)	50 (37%)	73	45.1%
2002	140	0%	0%	29 (20.7%)	111	20.7%
2003	138	4 (2.8%)	22 (15%)	65 (47.8%)	66	62.7%
2004	125	0%	4 (3.2%)	71 (56%)	50	60%

Figure 4 shows the monthly averaged temperature anomalies at 50 hPa for the latitudinal region 90° S – 65° S. Anomalies are departures from the 1979–99 base period mean. The figure is taken from the NOAA/ National Weather Center website (http://www.cpc.ncep.noaa.gov/products/analysis_monitoring/bulletin/figs3.html).

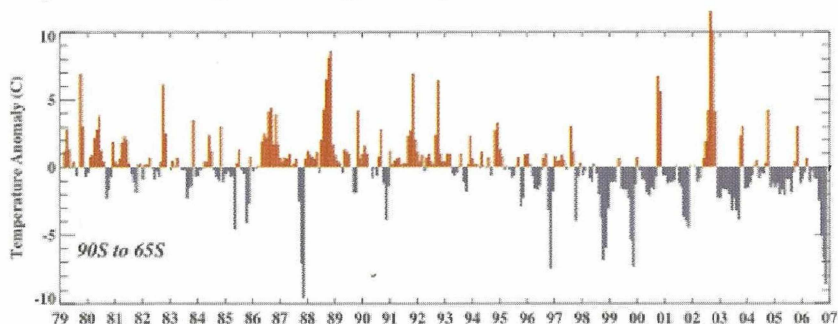


Fig. 4: Monthly averaged temperature anomalies at 50 hPa for the latitudinal region 65° S – 90° S.

Negative temperature anomaly up to approximately -2°C was observed during spring and winter of 2004. While, as compared to 2004, slight higher negative temperature up to anomaly -5°C was observed before and during ozone hole period, warming was seen in November and December 2003. However, unusual strong positive temperature anomaly up to 10°C magnitude was observed during ozone hole period in 2002. In 1997, positive temperature anomaly was observed quite early before ozone hole period followed by cooling in July-August and again warming in September-October. Temperature anomalies were found to be coinciding with observed variation in total ozone in 1997, 2002, 2003 and 2004. The magnitude of planetary waves and the heat flux transport toward the South Pole and upward transport of the wave energy was very strong in the year 2002 in southern hemisphere. In the year 1997, planetary wave activity was slightly strong; however, in the year 2003 and 2004, it was unusually weak during the formation of ozone hole in the month of August and early September. The anomalous variation of TOC has been observed over various stations at Antarctica during the ozone hole period of 1997 and 2002. In the year 2002, in an unusual event ozone hole split into two parts (as shown in Fig. 5) and showed an early recovery.

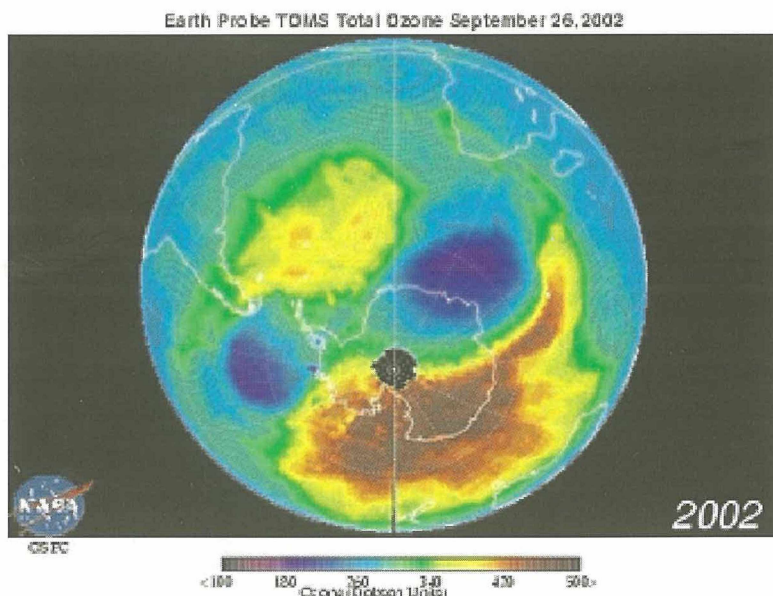


Fig. 5: Satellite imagery of ozone hole over Antarctica on 26 September 2002

However, the appearance of a deep ozone hole (126 DU), with longer duration, over Maitri in 2003 confirmed that the weaker ozone hole observed in the year 2002 does not denote a recovery of the ozone layer. Measurements show that the CFC concentrations in the stratosphere are levelling off. In the troposphere, CFC concentrations have started to decline due to the decisions taken by the international community under the Montreal Protocol, its subsequent adjustments and amendments to stop the use of ozone depleting chemicals. These measurements indicate that the ozone hole is not worsening, and may soon start to improve. However, this improvement is going to come very slowly. Further, changing atmospheric conditions, which modulates TOC over Antarctica, can complicate the task of identifying ozone hole recovery because a full explanation of meteorological anomalies observed over Antarctica is not yet available.

Direct solar UVB irradiance study using TUV model

The UVB irradiance measurements were carried out at Maitri, using a MICROTUPS II during the year 2004. It measures the irradiances centered at 305, 312 and 320 nm. Microtops-II observations presented in Figure 6 gives an idea about the stability of the instrument used for the measurement of irradiances at 305, 312 and 320 nm and ozone.

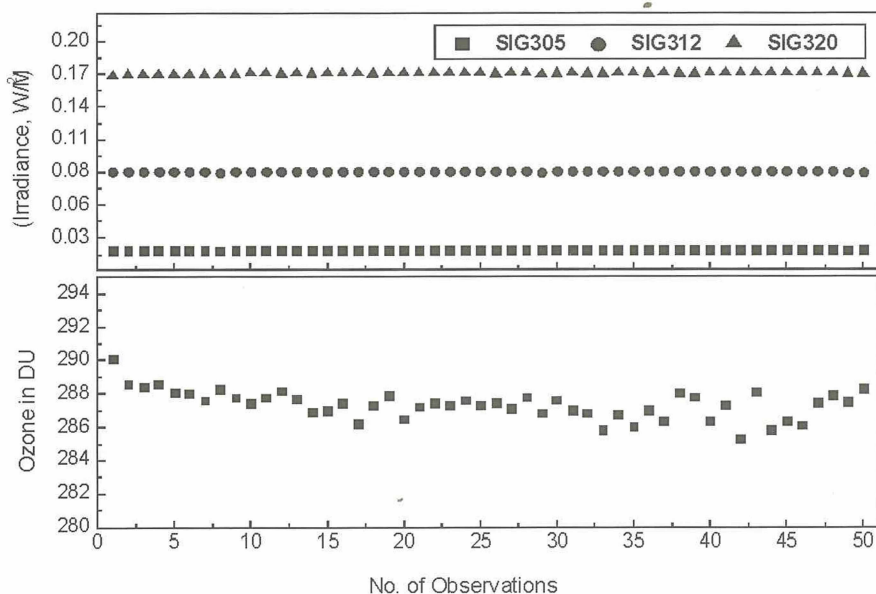


Fig. 6: Rapid measurements of column ozone (bottom) and direct solar irradiance (top) at 305, 312 and 320 nm wavelength done by MICROTUPS II Sun photometer

Fifty rapid observations of irradiances at 305, 312 and 320 nm each and ozone were made and standard deviation was calculated and is shown in Table 2. The repeatability of the data by this instrument is better than 0.3% for ozone and ~0.6% for irradiance.

Table 2 : Performance of the MICROTOPS II Sun photometer

Measured Parameters	Standard Deviation, %
Irradiance 305 nm	0.57
Irradiance 312 nm	0.23
Irradiance 320 nm	0.30
Ozone	0.30

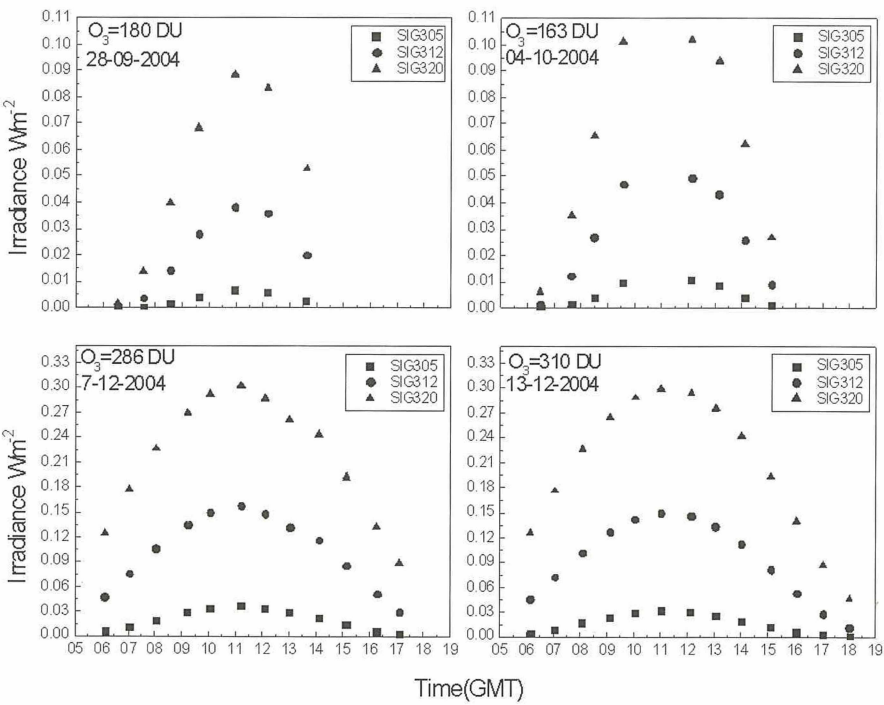


Fig. 7: Diurnal variation of clear-sky direct solar UV irradiance as observed during ozone hole period (top) and non-ozone hole period (bottom) for the three wavelengths 305, 312 and 320 nm. The corresponding dates and column ozone values are indicated in the legends.

Figure 7 demonstrates, for four different days, the complete daytime observations of direct solar irradiance values in W m^{-2} at 305, 312 and 320 nm. It shows the diurnal variation of UV-B irradiance during the non-ozone hole period on December 7 (ozone = 285 DU) and December 13 (ozone = 310 DU) 2004, and during the ozone hole period on September 28 (ozone = 180 DU) and October 04 (ozone = 163 DU), 2004. The maximum instantaneous UV-B irradiance observed at local noon during non-ozone hole period (13 December 2004, SZA = 47.59) was about 0.031, 0.150 and 0.299 W m^{-2} at the three wavelengths 305, 312 and 320 nm, respectively. The corresponding values during ozone hole period particularly on 4 October (SZA = 67.02) was about 0.010, 0.049 and 0.102 W m^{-2} respectively. This difference is mainly due to the large variation in SZA.

Tropospheric Ultraviolet and Visible (TUV) Radiation Model

Numerical modeling is an effective approach to study the UV variability and responsible mechanisms. The widely-used NCAR TUV model has been continuously updated and well documented. The model calculates the spectral irradiation, spectral actinic flux, photo dissociation coefficients (J-values), and biologically effective irradiance (dose and rates) over the wavelength range 121-750 nm. The TUV module takes into account the aerosol effect, which needs the input of aerosol optical depth. To study the UV variability, we have used latest TUV model version 4.2 to evaluate the measured UV irradiances under clear sky conditions. More details regarding this model can be found at <http://www.acd.ucar.edu/TUV> and Madronich (1993).

Comparison of Observed and TUV Modeled values

Figure 8 shows the comparison of the observed clear sky direct solar irradiance on December 13, 2004 with the TUV model estimated values at the three wavelengths 305 312 and 320 nm. It should be noted that the model calculations have been done with the average columnar ozone 310 DU measured by the instrument and aerosol optical depth set to 0.012. A comparison between the observation and TUV model curves clearly shows that both have similar diurnal characteristics, however, the irradiance calculated by the model at 320 nm is higher, and at 305 and 312 nm is lower than the observed values. The maximum differences between the clear-sky modeled and measured values are less than 15%, 8% and 6% for wavelengths 305, 312 and 320 nm, respectively on December 13, 2004 at Maitri. The close agreement between observed and modeled values (measured ozone values as an input to the model) indicates that that TUV model can be used to simulate UV irradiance for the same wavelengths at which MICROTOPS measures.

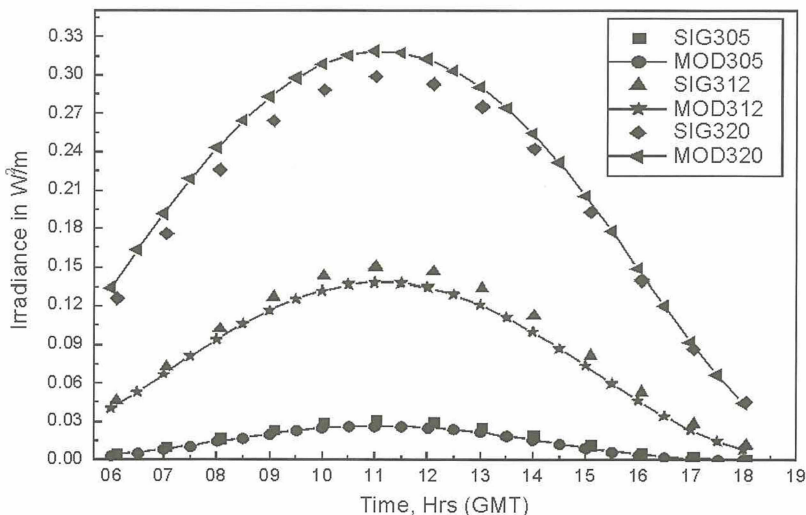


Fig. 8: Comparison of measured direct solar irradiance (dots) with the TUV model calculated (symbols with line) on December 13, 2004 for Maitri. The model calculations were done for average ozone (310 DU) and AOD (0.012)

Ozone-hole effect

In order to see the effect of ozone hole on UV irradiance incident at ground, the clear sky direct irradiance has been estimated using TUV model for about 10 days during ozone hole period (30 September to 10 October 2004) for two different values of column ozone. One, with the daily average ozone values observed during this ozone hole period and the other with the average ozone values that were found when there was no ozone hole over Antarctica (before 1980s), which is about 300 DU (300 DU is the minimum zonally averaged ozone over the higher southern latitude during August – October in 1979). The later value was kept constant at 300 DU for all ten days of calculations. I_{hole} represents the irradiance estimated by the TUV model for ozone hole period and I_{normal} is the irradiance estimated by TUV model assuming ozone values before ozone hole episodes over Antarctica. The difference between the two irradiances gives an idea of the change in irradiance during ozone-hole period due to loss of ozone. Table 3 shows the daily average ozone values measured by MICROTOS during the ozone hole period (30 September to 10 October 2004), and percentage loss of columnar ozone with respect to the normal ozone value of 300 DU.

Table 3 : Daily mean columnar ozone and percentage loss of ozone with respect to 300 DU over Maitri, Antarctica during 30 Sept – 10 Oct 2004

Date	9/30	10/1	10/2	10/3	10/4	10/5	10/6	10/7	10/8	10/10	Average
Ozone in DU	165.7	181.8	168.2	169.2	165.9	159.8	161.2	168.2	161.4	188.5	169
% Loss	44.77	39.40	43.93	43.60	44.70	46.73	46.27	43.93	46.20	37.17	43.7%

Figure 9 shows the daily mean irradiance corresponding to I_{hole} and I_{normal} obtained from the model for the corresponding days. A close observation of the Figure 9 reveals that the change in UV flux is maximum for 305 nm and minimum for 320 nm. The mean irradiance at 305, 312 and 320 nm corresponding to I_{hole} during the ozone hole period is found to be 0.0041, 0.023 and 0.060 Wm^{-2} . Similarly, for I_{normal} it is 0.00082, 0.0121 and 0.0485 Wm^{-2} at 305, 312 and 320 nm respectively. This indicates an increase of about 410%, 90% and 25% in UV irradiances at 305, 312 and 320 nm respectively. It is to be noted that this increase in UV irradiance corresponds

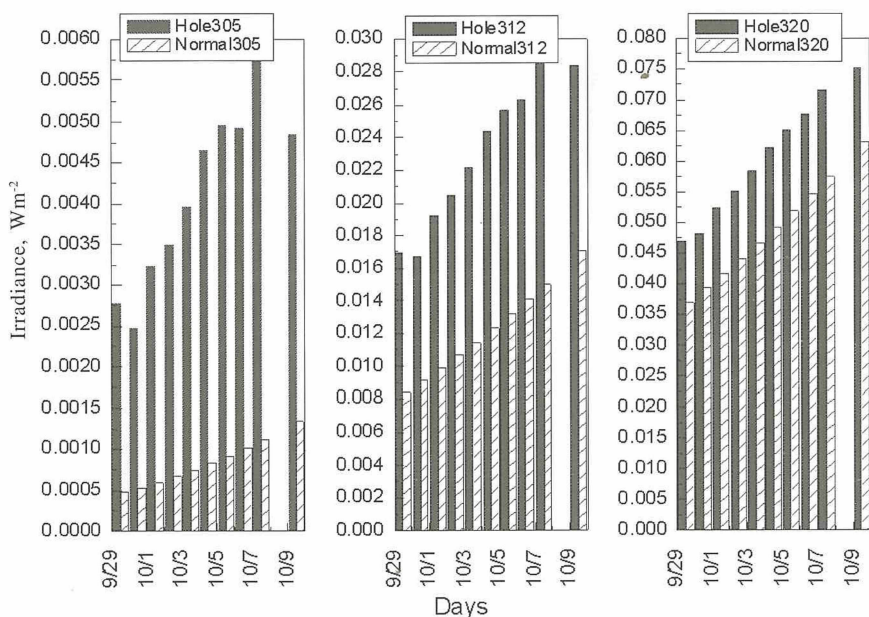


Fig. 9: A comparison of UV irradiance estimated using TUV model at 305 (left), 312 (center) and 320 (right) nm for the value of column ozone measured during ozone hole period (Sept. 30 – Oct. 10) in 2004 and with the normal average ozone column of 300DU.

to an average loss of ozone column to the order of 44% during the ozone hole period.

In order to see the change in irradiance due to the unit change in column ozone values at different solar zenith angles the TUV model was run several times for varying columnar ozone values and model irradiances were estimated for the three wavelengths. Keeping the zenith angle constant at 55°, 65° and 70°, the ozone parameter was varied from 320 DU to 100 DU in the steps of 20 DU. The change of direct irradiance as a function of total ozone is described by Beer Lambert's Law given by;

$$E = E_{o \text{ exp}} (-\alpha \mu x) * \tau_{\text{Ray}} * \tau_{\text{Aer}} \quad \dots (1)$$

where E_o is the irradiance outside the Earth's atmosphere, α is the ozone absorption coefficient, μ is the ratio of the actual and vertical path lengths through the ozone layer, x is total ozone and τ_{Ray} and τ_{Aer} are terms related to Rayleigh and aerosol scattering. At a given SZA and assuming other parameters as constant the UV irradiance will decrease exponentially with the column ozone. This is clearly indicated in the Fig. 10, which also shows the exponential fit on each set of data thus obtained. The statistics of exponential relationship between columnar ozone and UV irradiance for all the three wavelengths 305, 312 and 320 nm is illustrated in Table 4. The relative change of direct UV irradiance in response to a change in total ozone is always larger at larger SZA. This is a consequence of Beer Lambert's Law and the fact that μ increases with SZA.

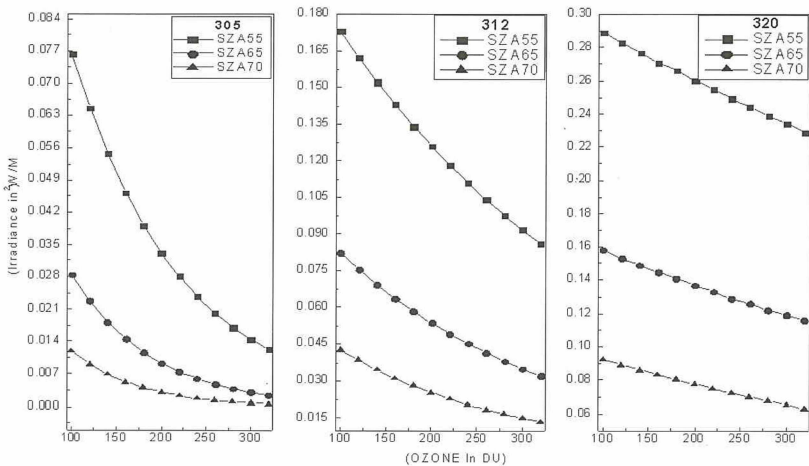


Fig. 10: Model Irradiance at varying ozone concentration (320-100 DU) for the wavelengths 305, 312 and 320 nm (lines with dot) keeping SZA constant at 55°, 65° and 70°. The lines represent the exponential fit on each set of data.

Table 4 : Exponential fit equation for different SZA for the wavelengths 305, 312 and 320 nm for the columnar ozone range of 320 – 100 DU.

	305 nm	312 nm	320 nm
SZA55	$y = 0.1747e^{-0.0083x}$	$y = 0.238e^{-0.0032x}$	$y = 0.3212e^{-0.0011x}$
SZA65	$y = 0.0863e^{-0.0111x}$	$y = 0.1261e^{-0.0043x}$	$y = 0.1818e^{-0.0014x}$
SZA70	$y = 0.0461e^{-0.0136x}$	$y = 0.0717e^{-0.0052x}$	$y = 0.1097e^{-0.0017x}$

It is also observed that the rate of change of irradiance per DU change in columnar ozone is higher at low zenith angles and decreases gradually as the zenith angle increases. The relationship between changes in solar irradiance per DU change in columnar ozone was investigated by TUV model.

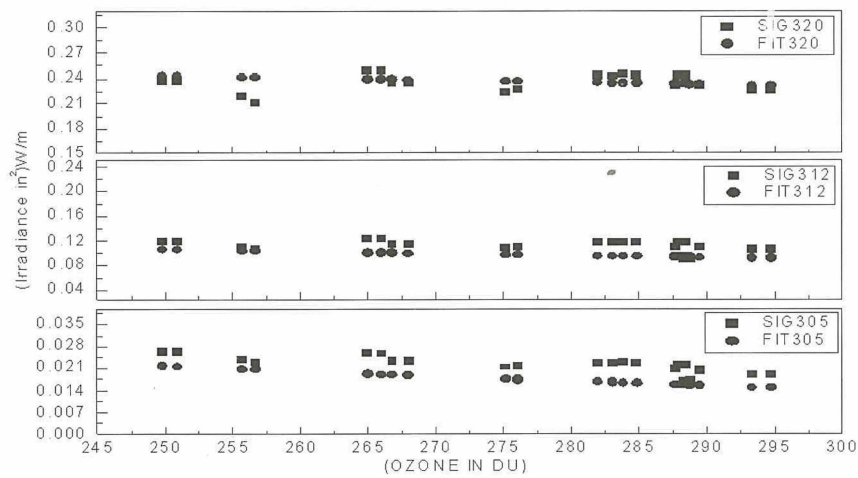


Fig. 11: Comparison between the direct clear sky solar irradiance measured by MICROTUPS during December 2004 and the irradiance obtained from the curve fit on modeled data set (Figure 6) for all the three wavelengths.

The relation obtained was also used to compare the observed values with the values obtained by curve fitting on the modeled set of data. These calculations were done for all the three wavelengths keeping the solar zenith angle constant at 55° and shown in Figure 11. It shows the comparison between the direct clear sky solar irradiance measured by MICROTUPS during December 2004 and the irradiances obtained from the curve fit on

modeled data set for all the three wavelengths. One can notice that the observed values match quite well with the estimated values from the exponential fit. Slight difference between the observed values and estimated values from the exponential fit is well within the standard deviation of the instrument given in Table 2.

CONCLUSION

Year-to-year variations in the depth and duration of the ozone hole over Schirmacher region of east Antarctica have been observed. Comparison with monthly averaged temperature anomalies at 50 hPa for the latitudinal region 65°S – 90°S reveals the direct relationship between the stratospheric temperature anomaly and variation in total ozone over the Antarctica. The day-to-day change of total ozone over Schirmacher region indicates that the chemical ozone destruction inside the polar vortex and duration of ozone hole is largely controlled by the dynamical behavior of the factors that influences the chemical ozone destruction. The abrupt anomalously high total ozone was observed during the ozone hole period of 1997 and 2002 over Schirmacher region. This was due to unstable polar vortex caused by sudden stratospheric warming resulting in the enhanced advection of ozone rich air into the polar vortex.

The instantaneous irradiance observed at local noon during the no-ozone-hole period (13 December 2003, $\text{SZA} = 47.59$) was about 0.031, 0.150 and 0.299 Wm^{-2} , whereas corresponding values during the ozone hole period (4 October, $\text{SZA} = 67.02$) was about 0.010, 0.049 and 0.102 Wm^{-2} at the three wavelengths 305, 312 and 320 nm, respectively. The entire clear-sky direct solar radiation measurements were compared with TUV model. The comparisons show a good agreement in characteristic variations of irradiances (W/m^2) with a maximum difference of about 14%, 8% and 7% at the peak values at wavelengths 305, 312, and 320 nm respectively. The effect of ozone hole on the clear sky direct UV irradiances incident at ground estimated using the TUV model suggests that for an average loss of ozone column by about 44% during the ozone hole period, the irradiance at the three wavelengths 305, 312, and 320 nm increases by about 403%, 88%, and 24% respectively. This is of particular importance at high latitudes such as Antarctica where the ozone loss persists for a very long period (almost 5-7 weeks) during the spring season. This further suggests that the TUV model can be used to estimate UV-B irradiance (280-320 nm) and earthemal irradiance at Antarctica; however, further experimental study is needed to reduce the uncertainties in the observations and model calculations.

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